

3.2.7 Factors such as size and metabolic rate affect the requirements of organisms and this gives rise to adaptations such as specialised exchange surfaces and mass transport systems.

Size and surface area	The relationship between the size of an organism or structure and surface area to volume ratio. Changes to body shape and the development of systems in larger organisms as adaptations that facilitate exchange as the ratio reduces. Candidates should be able to explain the significance of the relationship between size and surface area to volume ratio for the exchange of substances and of heat.
Gas exchange	Gas exchange • across the body surface of a single-celled organism • in the tracheal system of an insect (tracheae and spiracles) • across the gills of a fish (gill lamellae and filaments including the countercurrent principle) • by leaves of dicotyledonous plants (mesophyll and stomata). Candidates should be able to use their knowledge and understanding of the principles of diffusion to explain the adaptations of gas exchange surfaces. Structural and functional compromises between the opposing needs for efficient gas exchange and the limitation of water loss shown by terrestrial insects and xerophytic plants.
Mass transport	Over large distances, efficient supply of materials is provided by mass transport.
The blood system	The general pattern of blood circulation in a mammal. Names are required only of the coronary arteries and of blood vessels entering and leaving the heart, liver and kidneys. The structure of arteries, arterioles and veins in relation to their function. The structure of capillaries and their importance in metabolic exchange. The formation of tissue fluid and its return to the circulatory system.
The passage of water through a plant	The structure of a dicotyledonous root in relation to the pathway of water from root hairs through the cortex and endodermis to the xylem. Apoplastic and symplastic pathways. Transpiration and the effects of light, temperature, humidity and air movement. The roles of root pressure and cohesion tension in moving water through the xylem.